Frequency Tables

* Addition Law of Probability :-

It is used to find the probability that at least one of the two events occur.

Prob. of A or B occurring

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

* Multiplication Law of Probability :-

It is the probability of happening two events together.

$$P(A \cap B) = P(A) \times P(B)$$

Ex- On rolling a die → Let two events be-

Event A = Getting an odd no.

Event B = Getting a no. greater than 2

Subset for A → $\{1, 3, 5\}$

" " B → $\{3, 4, 5, 6\}$

a) Addition Law → $P(A) = \frac{3}{6}$, $P(B) = \frac{4}{6}$, $P(A \cap B) = \frac{2}{6}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{4}{6} - \frac{2}{6}$$

$$= \frac{3+4-2}{6}$$

$$P(A \cup B) = \frac{5}{6}$$

b) Multiplication Law → $P(A) = \frac{3}{6}$, $P(B) = \frac{4}{6}$

$$P(A \cap B) = P(A) \times P(B)$$

$$= \frac{\cancel{18}}{\cancel{6}_2} \times \frac{\cancel{4}^2}{\cancel{6}_3}$$

$$= \frac{1}{2} \times \frac{2}{3}$$

$$P(A \cap B) = \frac{2}{6}$$

* Marginal Probability :-

It is the probability of a single event without considering any other event.

Ex- Let the data of 100 students is taken

	Passed	Failed	Total
Male	30	20	50
Female	40	10	50
Total	70	30	100

CONTINGENCY TABLE

Here, 2 events are there

Event A → Gender

Event B → Result

Q.1 → Prob. that a randomly selected student is a Male.

Considering Gender & Ignoring Result

0.2 → Prob. that a student passes the exam.

Considering Result & Ignoring Gender

∴ Marginal Prob. tells us the overall prob. of one event by ignoring the other.

★ Joint Probability :-

It is the probability of two or more events happening together at the same time.

It is same as intersection of events

$P(A \cap B)$ ⇝ Joint Prob.

Ex- What is the prob. that a Male student is passed?

$$P(A \cap B) = \frac{30}{100} = 0.3$$

★ Conditional Probability :-

It is the prob of the occurrence of one event given second event has already occured

Formula →

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{——} \quad (1)$$

& vice versa

$$P(B|A) = \frac{P(A \cap B)}{} \quad \text{——} \quad (2)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{——— (2)}$$

Ex-
- Prob. of rain given clouds
- Prob. of India winning the match given Kohli scored a century
- Prob. of fog given season is winter

Ex- Drawing Cards →

Event A → Drawing a red card

$$P(A) = \frac{26}{52}$$

Event B → Drawing a face card

$$P(B) = \frac{\cancel{12}^{6}}{\cancel{52}_{26}}$$

$$P(A \cap B) = P(A) \times P(B)$$

$$= \frac{\cancel{26}^{1}}{\cancel{52}_{\cancel{2}_{1}}} \times \frac{\cancel{12}^{\cancel{6}^{3}}}{\cancel{52}_{26}}$$

$$= \frac{3}{26}$$

$$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{3/26}{6/26} = \frac{\cancel{3}^{1}}{\cancel{6}_{2}}$$

Conceptual Understanding → Prob (Red | face)

$$P(F) = \frac{12}{52} \quad ; \quad P(R|F) = \frac{\cancel{6}^{1}}{\cancel{12}_{2}} = \frac{1}{2}$$

* Baye's Rule / Baye's Theorem :-

Acc. to Conditional Prob →

From eq. (1)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

or $$P(A \cap B) = P(A|B) \cdot P(B) \quad \text{——— I}$$

From eq. (2)

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

or $$P(A \cap B) = P(B|A) \cdot P(A) \quad \text{——— II}$$

From I & II → LHS are same

So, comparing RHS

$$P(A|B) \times P(B) = P(B|A) \times P(A)$$

or

$$\boxed{P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}}$$

This is known as Baye's Rule.